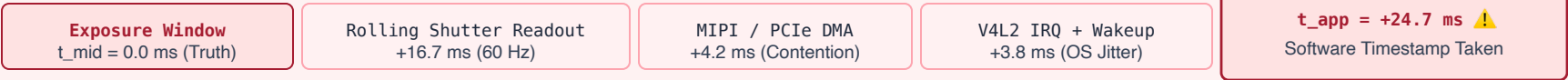


MULTI-SENSOR HARDWARE SYNCHRONIZATION & TIMESTAMP METROLOGY

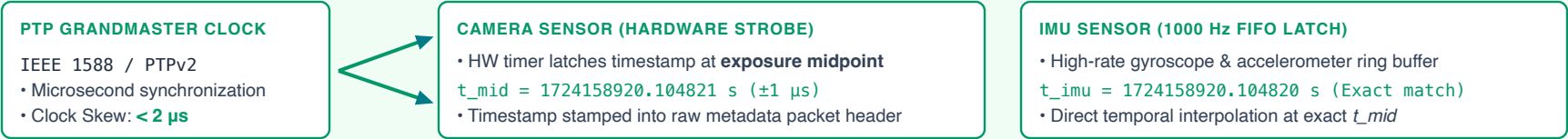
Eliminating cross-modal phase skew between camera photons and IMU dynamics via IEEE 1588 PTP hardware timestamps

✗ FLAWED: NAIVE SOFTWARE TIMESTAMPS (MEASURING APPLICATION ARRIVAL TIME)



Consequence for State Estimation: Kalman Filter / Factor Graph fuses camera pose from $t = 0.0 \text{ ms}$ with IMU velocity from $t = +24.7 \text{ ms}$.
At robot speed $v = 1.5 \text{ m/s}$, this $+24.7 \text{ ms}$ timestamp error injects **37.0 mm of fictitious kinematic error**, causing filter divergence and motor instability.

✓ PHYSICAL AI STANDARD: IEEE 1588 PTP HARDWARE EXPOSURE MIDPOINT TIMESTAMPS



Exact Metric Alignment: Even if Linux OS takes 30 ms to deliver the image tensor to Python, the system knows the exact physical photons were captured at t_{mid} .

- Phase Skew reduced from **24.7 ms** to **$< 0.005 \text{ ms}$ ($5 \mu\text{s}$)**.
- State estimation error drops from **37.0 mm** to **$< 0.01 \text{ mm}$** , enabling millimeter-precision grasping and rock-solid dynamic stability.